

Problem 33. Let $a > b > 0$. Prove that

$$\sqrt{2}a^3 + \frac{3}{ab - b^2} \geq 10.$$

When does equality hold?

Problem 34. Let a, b, c be non-zero real numbers such that

$$\begin{cases} a^2 + b + c = \frac{1}{a}, \\ b^2 + c + a = \frac{1}{b}, \\ c^2 + a + b = \frac{1}{c}. \end{cases}$$

Prove that at least two of a, b, c are equal.

Problem 35. Each square of $n \times n$ chessboard either contains a rook or is empty. Suppose that for any two rooks not threatening each other there is an empty square which is threatened by each of these rooks. Find the maximal possible number of rooks on the chessboard.

Problem 36. Let O be the circumcentre of an acute triangle ABC . A line perpendicular to AO intersects the line segments AC and AB at D and E , respectively. Let K be a point on BC which is not on the line AO . The line AK intersects the circumcircle of $\triangle ADE$ again at L . Let M be the point symmetric to A with respect to the line DE . Show that the points K, L, M, O are concyclic.